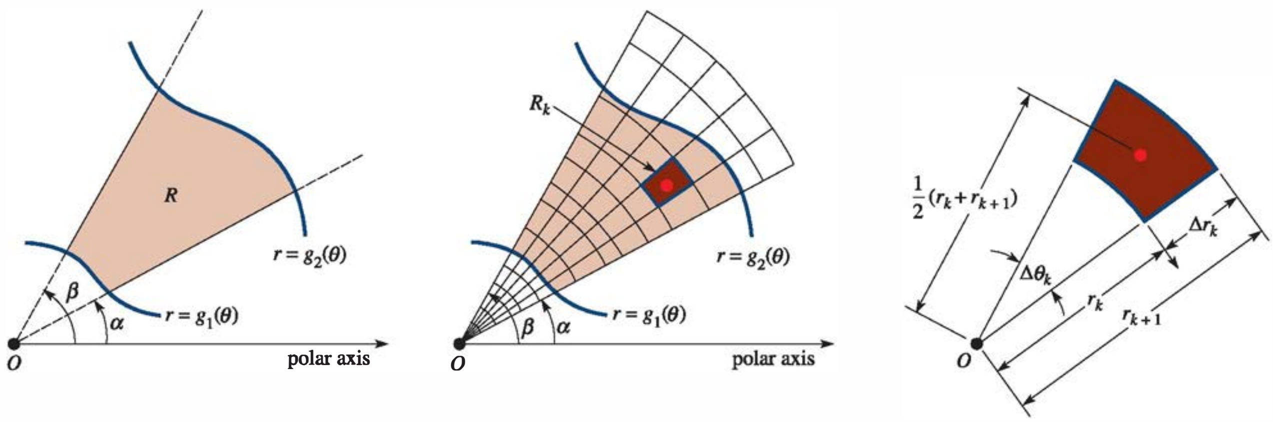
9.11 Double Integrals in Polar Coordinates

Introduction A double integral, which may be difficult or even impossible to evaluate in rectangular xy-coordinates, may become more tractable when expressed in a different coordinate system. In this section we examine double integrals in polar $r\theta$-coordinates.

Polar Rectangles Suppose R is a region in the plane bounded by the graphs of the polar equations $r = g_1(\theta)$, $r = g_2(\theta)$, and the rays $\theta = \alpha$, $\theta = \beta$, and f is a function of r and θ that is continuous on R. In order to define the double integral of f over R, we use rays and concentric circles to partition the region into a grid of **polar rectangles** or subregions R_k. See FIGURE 9.11.1(a) and 9.11.1(b). The area ΔA_k of typical subregion R_k, shown in Figure 9.11.1(c), is the difference of the areas of two circular sectors: $\Delta A_k = \frac{1}{2}r_{k+1}^2\Delta\theta_k - \frac{1}{2}r_k^2\Delta\theta_k$. Now ΔA_k can be written as

$$\Delta A_k = \tfrac{1}{2}(r_{k+1}^2 - r_k^2)\Delta\theta_k = \tfrac{1}{2}(r_{k+1} + r_k)(r_{k+1} - r_k)\Delta\theta_k = r_k^*\Delta r_k\Delta\theta_k$$



(a) Region R is bounded by polar graphs and rays

(b) Subregion R_k

(c) Enlargement of R_k

FIGURE 9.11.1 R_k in (b) and (c) is called a polar rectangle

where $\Delta r_k = r_{k+1} - r_k$ and $r_k^* = \frac{1}{2}(r_{k+1} + r_k)$ denotes the average radius. By choosing (r_k^*, θ_k^*) on each R_k, the double integral of f over R is

$$\lim_{\|P\|\to 0} \sum_{k=1}^{n} f(r_k^*, \theta_k^*) r_k^* \,\Delta r_k\, \Delta\theta_k = \iint_R f(r, \theta)\, dA.$$

The double integral is then evaluated by means of the iterated integral:

$$\iint_R f(r, \theta)\, dA = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r, \theta)\, r\, dr\, d\theta. \tag{1}$$

On the other hand, if the region R is as given in FIGURE 9.11.2, the double integral of f over R is then

$$\iint_R f(r, \theta)\, dA = \int_a^b \int_{h_1(r)}^{h_2(r)} f(r, \theta)\, r\, d\theta\, dr. \tag{2}$$

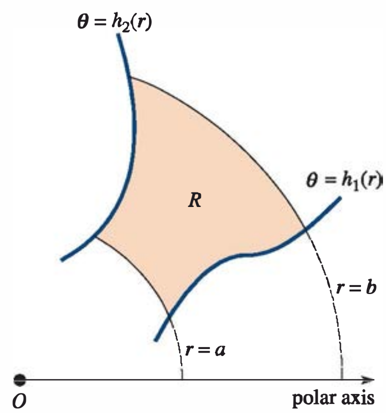


FIGURE 9.11.2 R bounded by polar graphs and circular arcs

EXAMPLE 1 Center of Mass

Find the center of mass of the lamina that corresponds to the region bounded by one leaf of the rose $r = 2\sin 2\theta$ in the first quadrant if the density at a point P in the lamina is directly proportional to the distance from the pole.

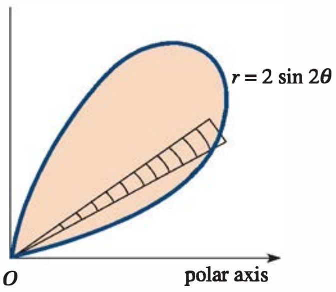


FIGURE 9.11.3 Lamina in Example 1

SOLUTION By varying θ from 0 to $\pi/2$, we obtain the graph in **FIGURE 9.11.3**. Now, $d(O, P) = |r|$. Hence, $\rho(r, \theta) = k|r|$, where k is a constant of proportionality. From (9) of Section 9.10, we have

$$m = \iint_R k|r|\, dA$$

$$= k\int_0^{\pi/2}\int_0^{2\sin 2\theta} (r)r\, dr\, d\theta$$

$$= k\int_0^{\pi/2} \frac{r^3}{3}\bigg]_0^{2\sin 2\theta} d\theta$$

$$= \frac{8}{3}k\int_0^{\pi/2} \sin^3 2\theta\, d\theta \qquad \leftarrow \sin^2 2\theta = 1 - \cos^2 2\theta$$

$$= \frac{8}{3}k\int_0^{\pi/2} (1 - \cos^2 2\theta)\sin 2\theta\, d\theta$$

$$= \frac{8}{3}k\left[-\frac{1}{2}\cos 2\theta + \frac{1}{6}\cos^3 2\theta\right]_0^{\pi/2} = \frac{16}{9}k.$$

Since $x = r\cos\theta$, we can write $M_y = k\iint_R x|r|\, dA$ as

$$M_y = k\int_0^{\pi/2}\int_0^{2\sin 2\theta} r^3\cos\theta\, dr\, d\theta$$

$$= k\int_0^{\pi/2} \frac{r^4}{4}\cos\theta\bigg]_0^{2\sin 2\theta} d\theta$$

$$= 4k\int_0^{\pi/2} \sin^4 2\theta\cos\theta\, d\theta \qquad \leftarrow \text{double angle formula}$$

$$= 4k\int_0^{\pi/2} 16\sin^4\theta\cos^4\theta\cos\theta\, d\theta$$

$$= 64k\int_0^{\pi/2} \sin^4\theta\cos^5\theta\, d\theta$$

$$= 64k\int_0^{\pi/2} \sin^4\theta(1 - \sin^2\theta)^2\cos\theta\, d\theta$$

$$= 64k\int_0^{\pi/2} (\sin^4\theta - 2\sin^6\theta + \sin^8\theta)\cos\theta\, d\theta$$

$$= 64k\left[\frac{1}{5}\sin^5\theta - \frac{2}{7}\sin^7\theta + \frac{1}{9}\sin^9\theta\right]_0^{\pi/2} = \frac{512}{315}k.$$

Similarly, by using $x = r\sin\theta$, we find*

$$M_x = k\int_0^{\pi/2}\int_0^{2\sin 2\theta} r^3\sin\theta\, dr\, d\theta = \frac{512}{315}k.$$

*We could have argued to the fact that $M_x = M_y$ and hence $\bar{x} = \bar{y}$ from the fact that the lamina and the density function are symmetric about the ray $\theta = \pi/4$.

Here the rectangular coordinates of the center of mass are

$$\bar{x} = \bar{y} = \frac{512k/315}{16k/9} = \frac{32}{35}.$$

Change of Variables: Rectangular to Polar Coordinates

In some instances a double integral $\iint_R f(x, y)\, dA$ that is difficult or even impossible to evaluate using rectangular coordinates may be readily evaluated when a change of variables is used. If we assume that f is continuous on the region R and if R can be described in polar coordinates as $0 \le g_1(\theta) \le r \le g_2(\theta)$, $\alpha \le \theta \le \beta$, $0 < \beta - \alpha \le 2\pi$, then

$$\iint_R f(x, y)\, dA = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r\cos\theta, r\sin\theta) r\, dr\, d\theta. \tag{3}$$

Equation (3) is particularly useful when f contains the expression $x^2 + y^2$ since, in polar coordinates, we can now write

$$x^2 + y^2 = r^2 \quad \text{and} \quad \sqrt{x^2 + y^2} = r.$$

EXAMPLE 2 Changing an Integral to Polar Coordinates

Use polar coordinates to evaluate

$$\int_0^2 \int_x^{\sqrt{8-x^2}} \frac{1}{5 + x^2 + y^2}\, dy\, dx.$$

SOLUTION From $x \le y \le \sqrt{8 - x^2}$, $0 \le x \le 2$, we have sketched the region R of integration in FIGURE 9.11.4. Since $x^2 + y^2 = r^2$, the polar description of the circle $x^2 + y^2 = 8$ is $r = \sqrt{8}$. Hence, in polar coordinates, the region of R is given by $0 \le r \le \sqrt{8}$, $\pi/4 \le \theta \le \pi/2$. With $1/(5 + x^2 + y^2) = 1/(5 + r^2)$, the original integral becomes

$$\begin{aligned}
\int_0^2 \int_x^{\sqrt{8-x^2}} \frac{1}{5 + x^2 + y^2}\, dy\, dx &= \int_{\pi/4}^{\pi/2} \int_0^{\sqrt{8}} \frac{1}{5 + r^2} r\, dr\, d\theta \\
&= \frac{1}{2} \int_{\pi/4}^{\pi/2} \int_0^{\sqrt{8}} \frac{2r\, dr}{5 + r^2} d\theta \\
&= \frac{1}{2} \int_{\pi/4}^{\pi/2} \ln(5 + r^2)\Big]_0^{\sqrt{8}} d\theta \\
&= \frac{1}{2}(\ln 13 - \ln 5) \int_{\pi/4}^{\pi/2} d\theta \\
&= \frac{1}{2}(\ln 13 - \ln 5)\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{\pi}{8} \ln \frac{13}{5}.
\end{aligned}$$

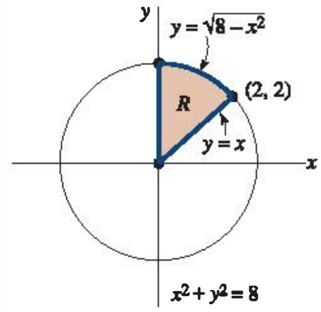


FIGURE 9.11.4 Region R in Example 2

EXAMPLE 3 Volume

Find the volume of the solid that is under the hemisphere $z = \sqrt{1 - x^2 - y^2}$ and above the region bounded by the graph of the circle $x^2 + y^2 - y = 0$.

SOLUTION From FIGURE 9.11.5, we see that $V = \iint_R \sqrt{1 - x^2 - y^2}\, dA$. In polar coordinates the equations of the hemisphere and the circle become, respectively, $z = \sqrt{1 - r^2}$ and $r = \sin\theta$. Now, using symmetry, we have

$$V = \iint_R \sqrt{1 - r^2}\, dA = 2\int_0^{\pi/2} \int_0^{\sin\theta} (1 - r^2)^{1/2} r\, dr\, d\theta$$

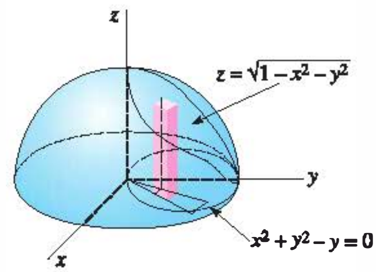


FIGURE 9.11.5 Solid in Example 3

$$2\int_0^{\pi/2}\left[-\frac{1}{3}(1-r^2)^{3/2}\right]_0^{\sin\theta} d\theta$$

$$\frac{2}{3}\int_0^{\pi/2}[1-(1-\sin^2\theta)^{3/2}]\ d\theta$$

$$\frac{2}{3}\int_0^{\pi/2}[1-(\cos^2\theta)^{3/2}]\ d\theta$$

$$\frac{2}{3}\int_0^{\pi/2}[1-\cos^3\theta]\ d\theta$$

$$\frac{2}{3}\int_0^{\pi/2}[1-(1-\sin^2\theta)\cos\theta]\ d\theta$$

$$\frac{2}{3}\left[\theta-\sin\theta+\frac{1}{3}\sin^3\theta\right]_0^{\pi/2} \quad \frac{\pi}{3}-\frac{4}{9}\approx 0.60. \qquad \equiv$$

Area Note that in (1) if $f(r, \theta)$ $\quad$ 1, then the **area** of the region R in Figure 9.11.1(a) is given by

$$A=\iint_R dA=\int_\alpha^\beta\int_{g_1(\theta)}^{g_2(\theta)} r\,dr\,d\theta.$$

The same observation holds for (2) and Figure 9.11.2 when $f(r, \theta)$ $\quad$ 1.

Remarks

The reader is invited to reexamine Example 3. The graph of the circle r $\quad$ $\sin\theta$ is obtained by varying θ from 0 to π. However, carry out the iterated integration

$$V \quad \int_0^{\pi}\int_0^{\sin\theta}(1-r^2)^{1/2} r\,dr\,d\theta$$

and see if the result is the *incorrect* answer $\pi/3$. What goes wrong?

9.11 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–4, use a double integral in polar coordinates to find the area of the region bounded by the graphs of the given polar equations.

1. r $\quad$ $3 + 3\sin\theta$
2. r $\quad$ $2 + \cos\theta$
3. r $\quad$ $2\sin\theta$, r $\quad$ 1, common area
4. r $\quad$ $8\sin 4\theta$, one petal

In Problems 5–10, find the volume of the solid bounded by the graphs of the given equations.

5. One petal of r $\quad$ $5\cos 3\theta$, z $\quad$ 0, z $\quad$ 4
6. $x^2 + y^2$ $\quad$ 4, z $\quad$ $\sqrt{9 - x^2 - y^2}$, z $\quad$ 0
7. Between $x^2 + y^2$ $\quad$ 1 and $x^2 + y^2$ $\quad$ 9, z $\quad$ $\sqrt{16 - x^2 - y^2}$, z $\quad$ 0
8. z $\quad$ $\sqrt{x^2 + y^2}$, $x^2 + y^2$ $\quad$ 25, z $\quad$ 0
9. r $\quad$ $1 + \cos\theta$, z $\quad$ y, z $\quad$ 0, first octant
10. r $\quad$ $\cos\theta$, z $\quad$ $2 + x^2 + y^2$, z $\quad$ 0

In Problems 11–16, find the center of mass of the lamina that has the given shape and density.

11. r $\quad$ 1, r $\quad$ 3, x $\quad$ 0, y $\quad$ 0, first quadrant; $\rho(r, \theta)$ $\quad$ k (constant)
12. r $\quad$ $\cos\theta$; density at point P directly proportional to the distance from the pole
13. y $\quad$ $\sqrt{3}x$, y $\quad$ 0, x $\quad$ 3; $\rho(r, \theta)$ $\quad$ r^2
14. r $\quad$ $4\cos 2\theta$, petal on the polar axis; $\rho(r, \theta)$ $\quad$ k (constant)
15. Outside r $\quad$ 2 and inside r $\quad$ $2 + 2\cos\theta$, y $\quad$ 0, first quadrant; density at a point P inversely proportional to the distance from the pole
16. r $\quad$ $2 + 2\cos\theta$, y $\quad$ 0, first and second quadrants; $\rho(r, \theta)$ $\quad$ k (constant)

In Problems 17–20, find the indicated moment of inertia of the lamina that has the given shape and density.

17. r $\quad$ a; $\rho(r, \theta)$ $\quad$ k (constant); I_x